

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-S4: Bivariate Data Analysis

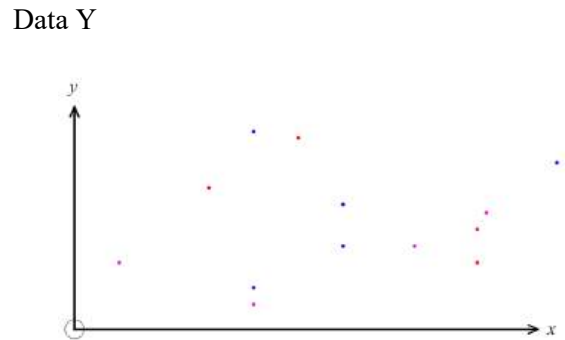
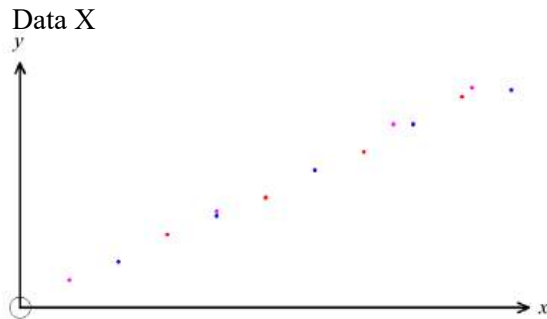
Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

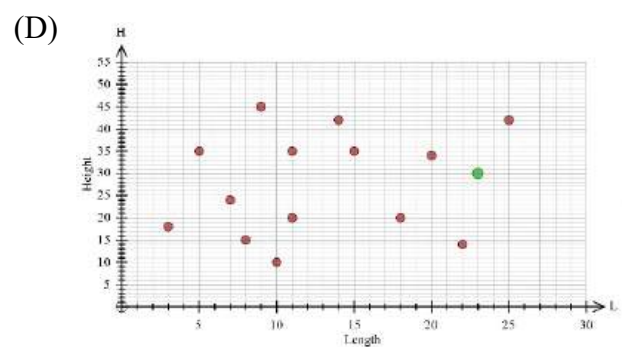
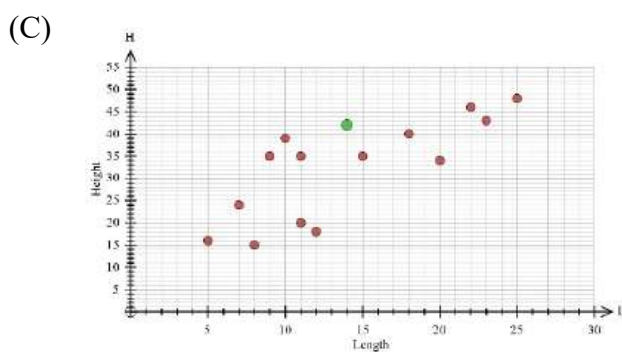
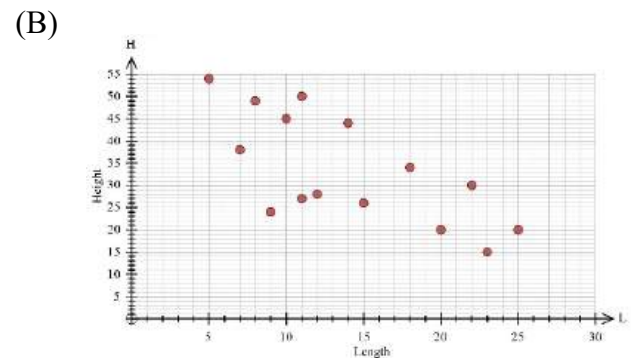
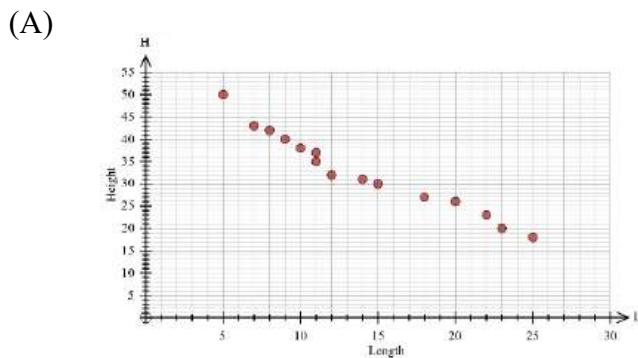
Section I

15 marks

1. Which best describes the two sets of data X and Y, shown below.

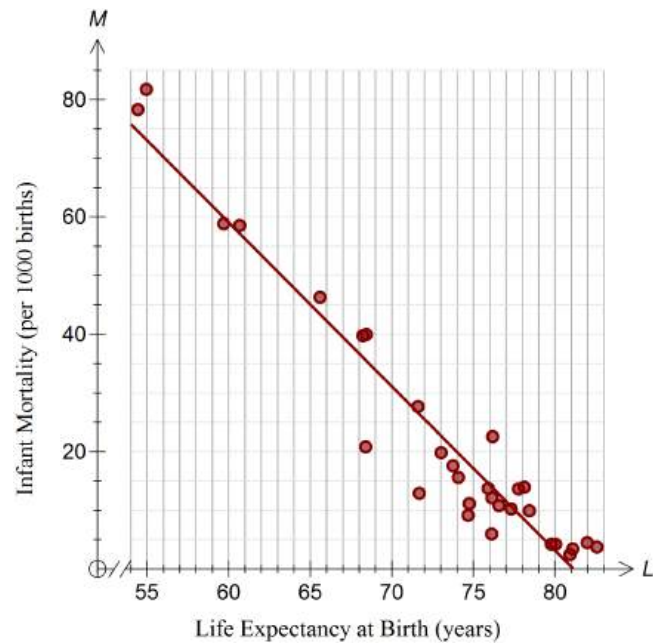


- (A) X has a strong negative correlation and Y has no correlation.
- (B) X has a strong negative correlation and Y has a strong positive correlation.
- (C) X has a strong positive correlation and Y has a strong negative correlation.
- (D) X has a strong positive correlation and Y has no correlation.
2. For a set of bivariate data, Pearson's correlation coefficient is -0.7 . Which graph could best represent this set of bivariate data?



Questions 3 and 4 refer to the following:

Josh researches data on the life expectancy at birth and the infant mortality rate for a number of countries. He then draws a scatterplot with a line of best fit.



3. What is the likely correlation coefficient for this data?

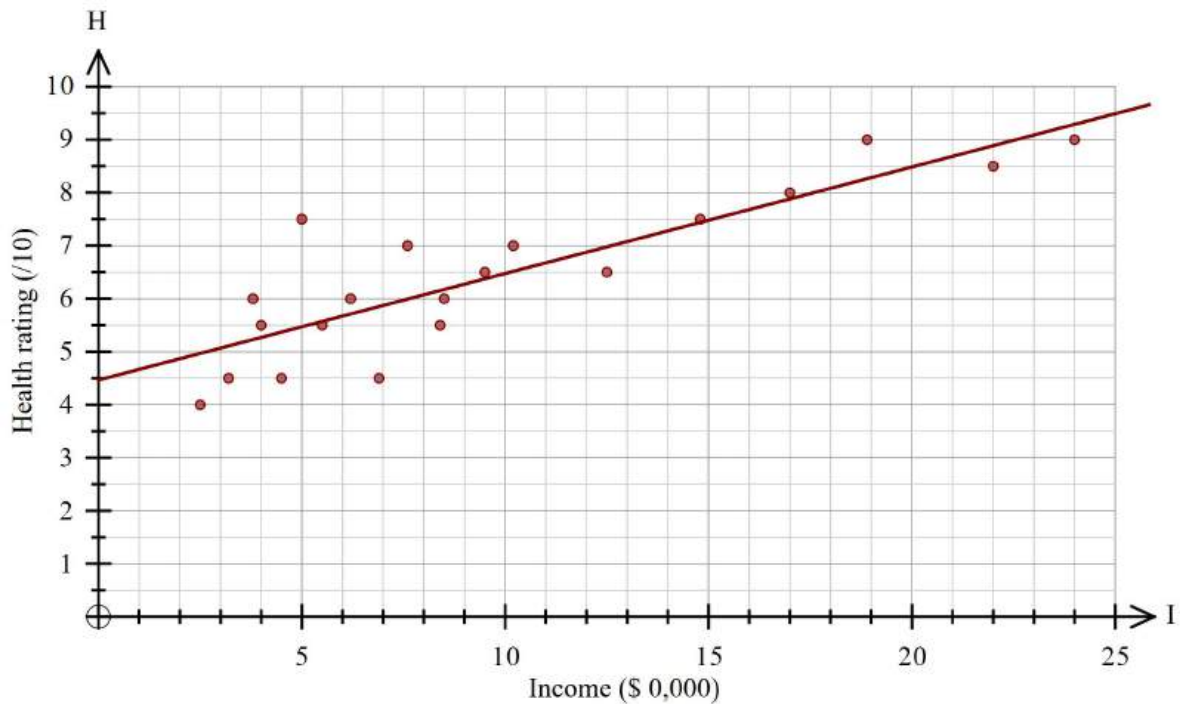
- (A) -0.95
- (B) -0.59
- (C) 0.59
- (D) 0.95

4. Which could be the equation of the regression line?

- (A) $M = 80 - 2.8L$
- (B) $M = 80 + 2.8L$
- (C) $M = 227 - 2.8L$
- (D) $M = 227 + 2.8L$

Questions 5 – 7 refer to the scatterplot below, which has a least-squares regression line drawn.

The mean income ($\bar{I}$) was \$97 500 and the point ($\bar{I}$, $\bar{H}$) lies on the regression line.



5. Which description could apply to the relationship between the two sets of data?
- (A) A weak negative correlation.
 - (B) A weak positive correlation.
 - (C) A moderately strong negative correlation
 - (D) A moderately strong positive correlation
6. What is the mean health rating?
- (A) 6.1
 - (B) 6.4
 - (C) 6.8
 - (D) 7.1
7. Which equation could describe the regression line?
- (A) $H = 0.2I - 4.5$
 - (B) $H = 0.2I + 4.5$
 - (C) $H = 2I - 4.5$
 - (D) $H = 2I + 4.5$

8. Jo draws a scatterplot to compare Hand Span (s) and Reaction Time (t) and notices a linear relationship. She calculates the Pearson's correlation coefficient as $r = -0.8$.

She also calculates the results below for the means and standard deviations.

	Horizontal (s) axis	Vertical (t) axis
	Hand Span (mm)	Reaction Time (ms)
Mean	165	820
SD	25	324

She adds a least-squares regression line to the scatterplot.

The equation of this line is $t = 2530 + ms$, where m is the slope of the regression line.

The least-squares regression line passes through the point $(\bar{s}, \bar{t})$.

where $\bar{s}$ is the sample mean of the hand span data and $\bar{t}$ is the sample mean of the reaction time data.

What is the gradient of the least-squares line of best fit (correct to 3 s.f.)?

- (A) -10.4
- (B) -4.0
- (C) 4.0
- (D) 10.4

Western Mathematics
Mathematics Standard 2
MS-S4: Bivariate Data Analysis
TOPIC REVISION QUESTIONS - Section II

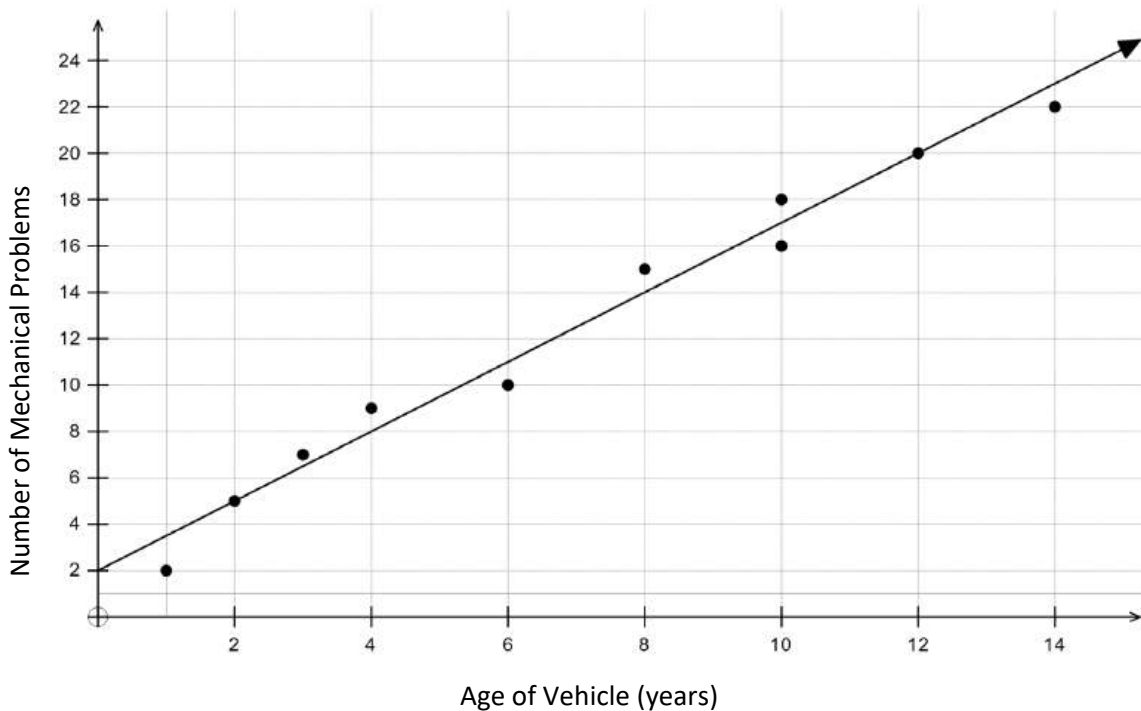
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (4 marks)

Marks

The scatter plot below compares the age of a vehicle with the number of mechanical problems it has experienced. A least-squares regression line has been drawn.



(a) Describe the correlation shown on the graph.

1

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(b) What is the gradient of the regression line?

1

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(c) What is the equation of the regression line?

1

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(d) Use the line to predict the number of problems experienced by a 20 year-old car.

1

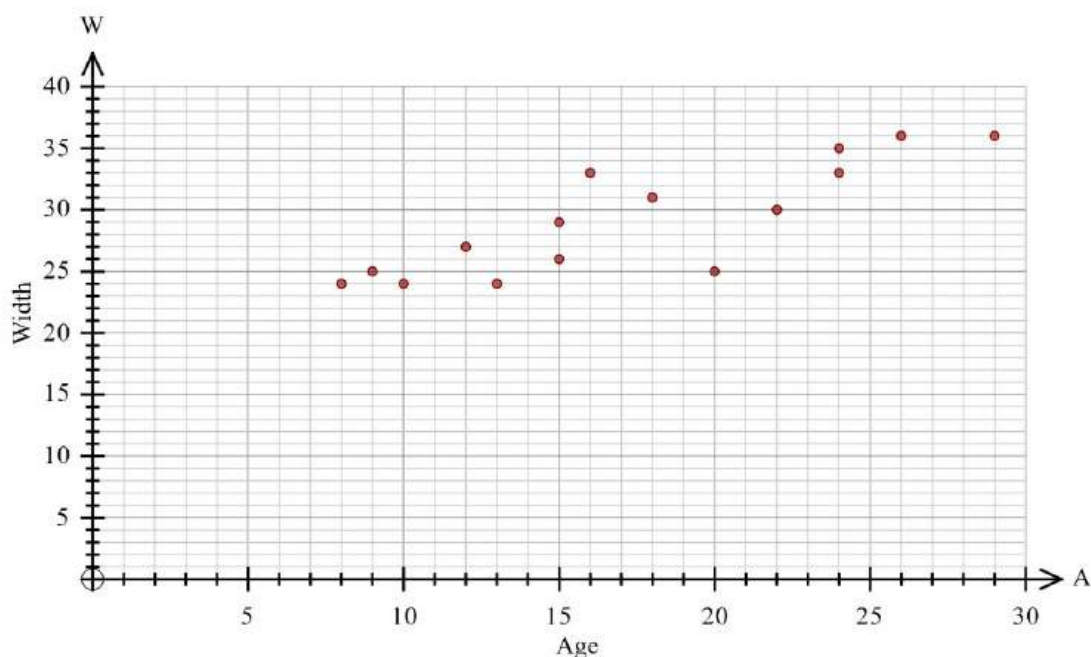
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Question 10 (5 marks)

Greg collects bivariate data and displays it in the scatterplot below.



The mean ($\bar{W}$) of the width data is 29.2.

- (a) Calculate the value of Pearson's correlation co-efficient for the bivariate data. 1

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- (b) She determines the equation of the least-squares regression line is $W = 0.59A + 19$. 1

Draw this line on the scatterplot.

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- (c) Use the equation to estimate the width for an age of 17. 1

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- (d) The least-squares regression line passes through $(\bar{A}, \bar{W})$. 1

Use the equation to estimate the value of $\bar{A}$.

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- (e) Explain why the equation should not be used to estimate the width for an age of 31. 1

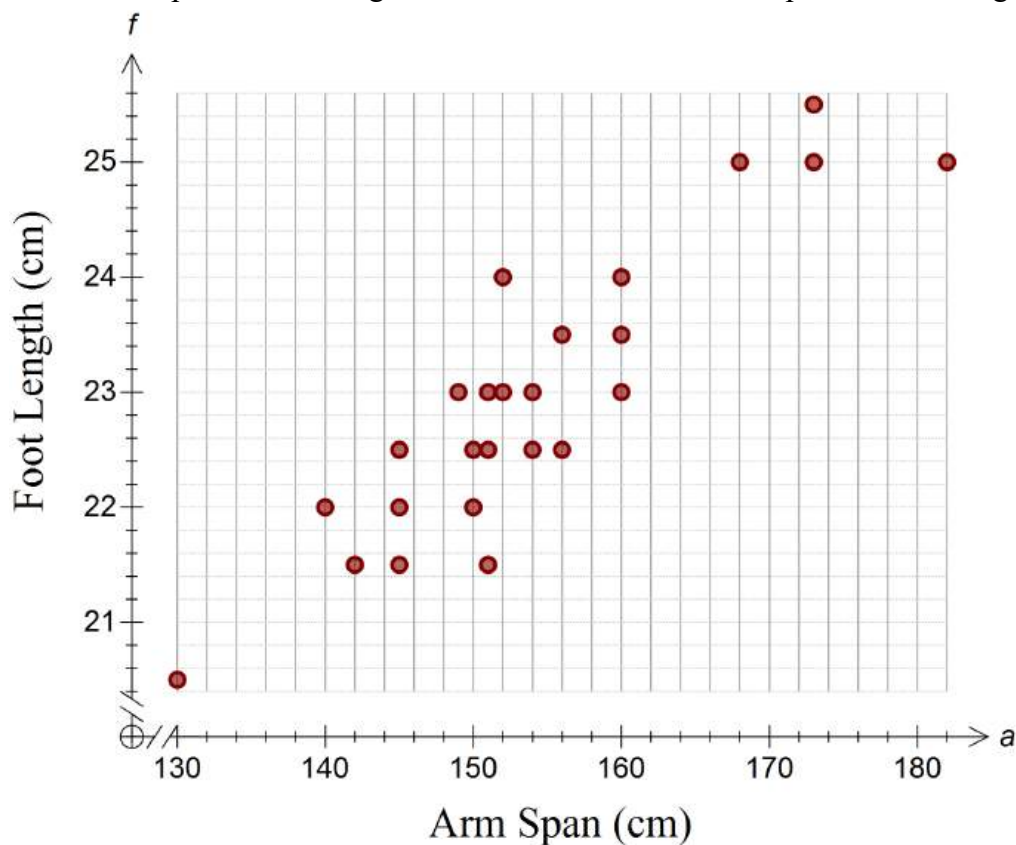
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Question 11 (6 marks)

Emily collects data from a sample of students in all years of her school.

She draws the scatterplot below using the data she collected on arm span and foot length.



- (a) Emily decides to include herself in the data. Her arm span was 164 cm and her foot length was 24 cm. 1
Plot a point on the graph to represent Emily and label it E.
- (b) Emily uses a statistics software package to calculate the correlation coefficient and gets a value of 0.9. 1
Explain what this result tells you about arm span and foot length.

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- (c) The statistics package also gives the results below for the mean and standard deviation of the two quantities.

2

	Arm Span (a)	Foot Length (f)
Mean	154.0	23.0
SD	11.3	1.3

Emily calculated that the gradient of the least-squares regression line is 0.1 (correct to 1 decimal place).

This regression line passes through the point $(\bar{a}, \bar{f})$.

Using the variables a and f , write down the equation of the least-squares regression line.

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- (d) Use the equation to estimate the foot length of a student whose arm span was 170 cm.

1

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- (e) Draw the least-squares regression line on the graph.

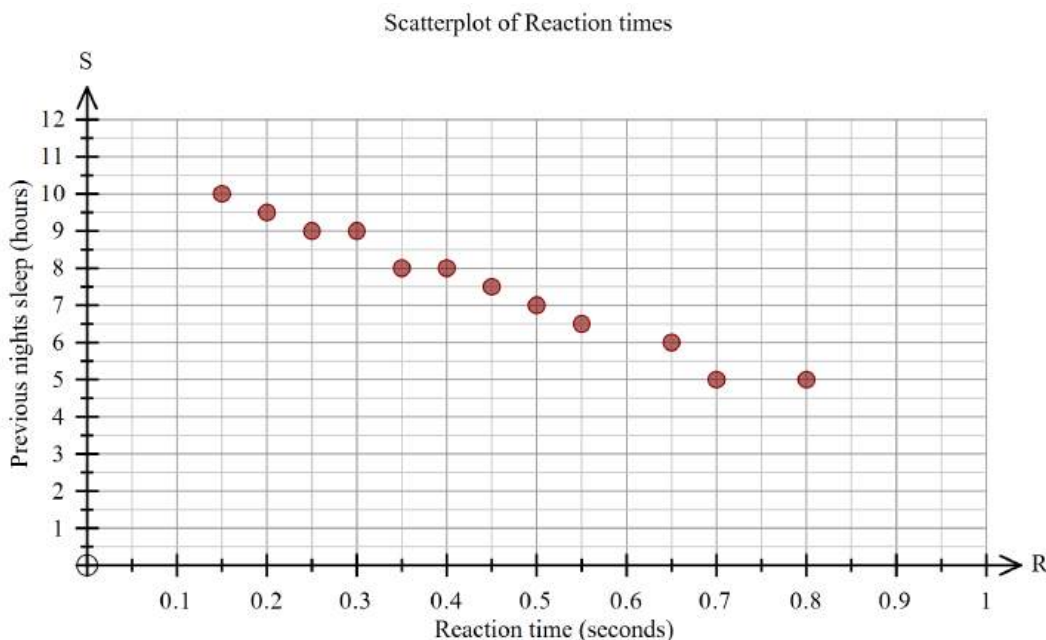
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Question 12 (4 marks)

Miriam was investigating the effect of lack of sleep on drivers.

She collected data on reaction times and compared these to the hours of sleep of the participants.

The results are shown on the scatterplot below.



- (a) Calculate Pearson correlation coefficient for the data.

1

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- (b) Describe the direction and the strength of the linear association between Reaction time and sleep.

1

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- (c) Miriam calculates the least squares line of best fit to have an equation of:

1

$$S = -8R + 11.$$

Draw the line on the graph above.

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- (d) Calculate the predicted sleep hours of a person who had a reaction time of 0.6 seconds.

1

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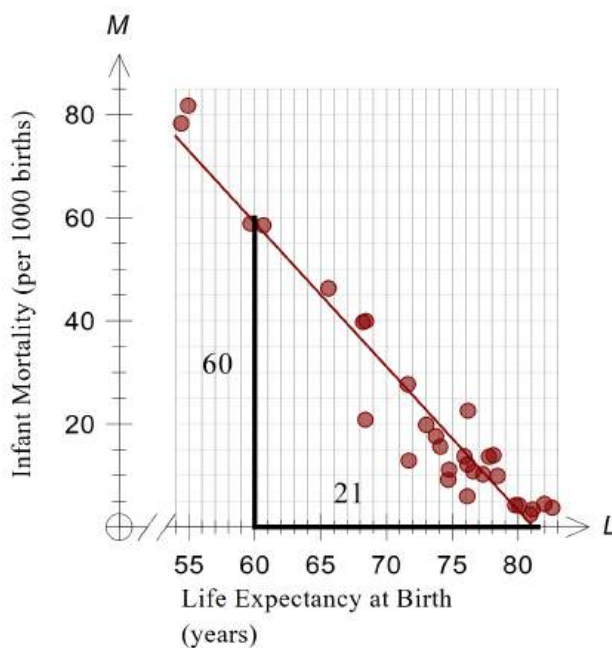
End of Revision

Western Mathematics MS-S4: Bivariate Data Analysis
Section I - Worked Solutions

Revision

No	Working	Answer
1	Data X approximates a straight line with a positive gradient, so is a strong positive correlation Data Y does not follow any linear or curved pattern, so there is no correlation	D
2	A negative correlation indicates as one increases, the other decreases, or the trend goes down from left to right, so only A and B meet this. (C is a moderately strong positive correlation and D is a weak positive correlation) A correlation of -0.7 indicates a moderately strong relationship, so B is most likely $r = -0.7$. A is a very strong relationship close to $r = -1$	B
3	One variable decreases as the other increases, so the correlation will be negative. The data is grouped close to the line so the value will be close to -1. So -0.95 is the most likely.	A

4



Gradient = $m = -\frac{60}{21} \approx -2.857$ (Take 2.8 from those given)

y – intercept is not 80 as there is a broken scale on horizontal axis

Could estimate it will be much higher than 80,

as it still has 55 horizontal units to reach vertical axis 4.5

Or using: $y = mx + b$

$M = -2.8L + b$ and substitute (60, 60) approx point on the line

$$60 = -2.8(60) + b$$

$$60 = -168 + b$$

$$b \approx 60 + 168 = 228 \text{ (nearest is 227 from those given)}$$

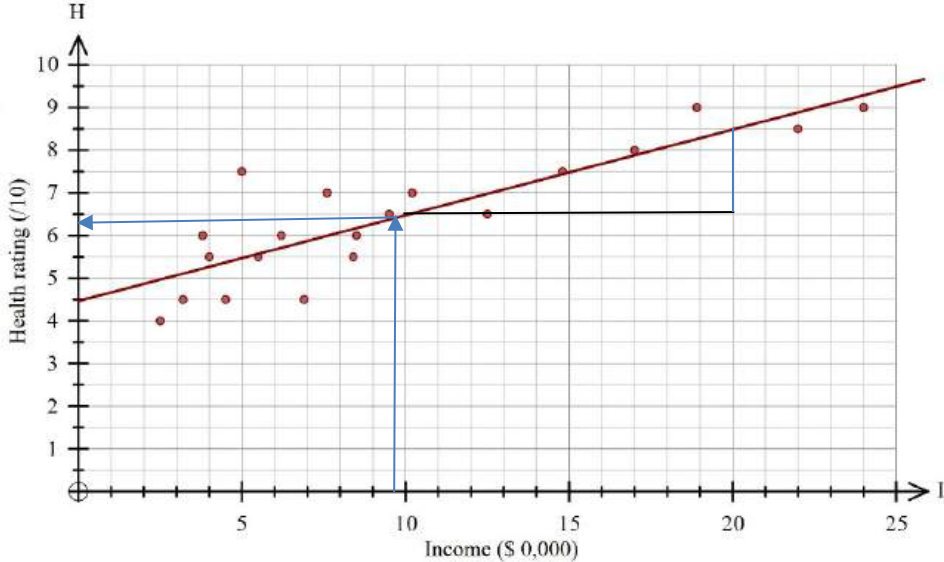
So $M = 227 - 2.8L$ is likely equation

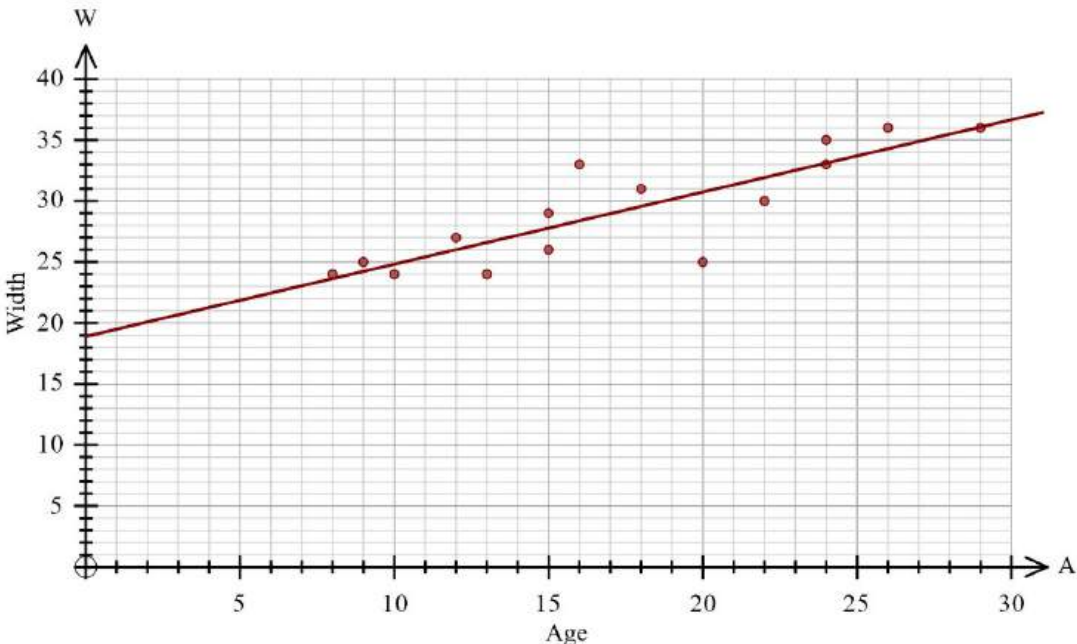
C

5

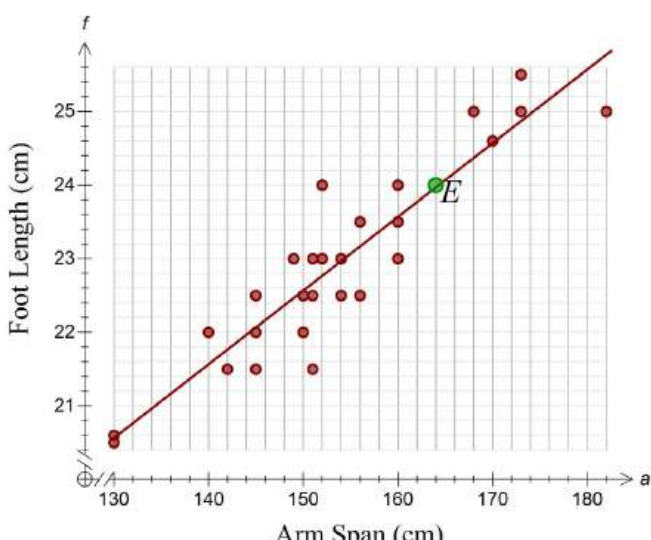
The line has a positive gradient so it is a positive correlation. The points are grouped reasonably closely around the line, so it is a moderately strong positive correlation.

D

6	 Since the point $(\bar{I}, \bar{H})$ lies on the regression line, the mean health rating can be found by reading off the mean income (97 500 becomes 9.75 on the scale) and following up to the regression line and reading across to the Health rating, obtaining 6.4.	B
7	Drawing a triangle through any two convenient points on the line, in this case we get a rise of $8.5 - 6.5 = 2.0$ and a run of $20 - 10 = 10$ $\text{Gradient} = m = \frac{2}{10} = 0.2$ y – intercept = 4.5 Using: $y = mx + b$ $H = 0.2I + 4.5$	B
8	$(\bar{s}, \bar{t}) = (165, 820)$ Substitute this into $t = 2530 + ms$ $820 = 2530 + 165m$ $165m = 820 - 2530$ $165m = -1710$ $m = -\frac{1710}{165}$ $= -10.36363636$ $= -10.4 \text{ (3 s f)}$	A

MS-S4: Bivariate Data Analysis		Marks
Section II Solution		
Question 9		
(a)	A strong positive correlation.	1
(b)	Gradient = $\frac{12}{8} = 1.5$	1
(c)	$N = 1.5Y + 2$ or equivalent with other variables.	1
(d)	$N = 1.5 \times 20 + 2 = 32$ problems	1
Question 10		
(a)	Using a calculator, $r = 0.85$ (to 2 sig fig).	1
(b)		1
(c)	$W = 0.59A + 19$ $A = 17$ $W = 0.59 \times 17 + 19$ $= 29.03$	1
(d)	$\bar{W} = 0.59\bar{A} + 19$ $\bar{W} = 29.2$ $29.2 = 0.59 \times \bar{A} + 19$ $29.2 - 19 = 0.59 \times \bar{A} = 10.2$ $\bar{A} = \frac{10.2}{0.59}$ $= 17.28813559$	1
(e)	As an age of 3 is well outside of the given scores, it would be extrapolating beyond the observed values which can be unreliable.	1

Question 11

(a)	 See point E on graph.	1
(b)	A correlation approaching 1 means that as one quantity increases, so does the other in a close linear relationship.	1
(c)	Gradient $m = 0.1$ Equation is of the form $f = 0.1a + b$ Substitute $(\bar{a}, \bar{f}) = (154.0, 23.0)$ $23.0 = 0.1 \times 154.0 + b$ $23.0 = 15.4 + b$ $b = 23.0 - 15.4 = 7.6$ Equation is $f = 0.1a + 7.6$	2
(d)	$f = 0.1a + 7.6$ Where $a = 170$, $f = 0.1 \times 170 + 7.6$ $= 24.6$	1
(e)	Plotting any 2 points from the equation in part (v) and joining to make a straight line. (See line in graph in part (a).)	1

Question 12

(a)	From calculator, $r = -0.99$ (2 sig fig).	1
(b)	Given the r value is close to negative one, it is a strong negative relationship.	1

(c)	Scatterplot of Reaction times Using y-intercept of 11 as starting point, go across 1, (to end of grid) and down 8. Or use equation to find any two or three points and plot line through these.	1
(d)	$S = -8R + 11$ $S = -8(0.6) + 11 = 6.2$	1